ANY FINITE TYPE CLOSED SUBSCHEME OF A QUASI-COMPACT QUASI-SEPARATED SCHEME SITS IN AN ALMOST BLOWUP SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EVA (sources/stacks-flat/pinned-original-source.tex line 11087).

Allowed standard prerequisites (editorial).

- Absolute Noetherian approximation. Limits, Proposition labelled limits-proposition-approximate, cited at proof lines 11094-11096. Writes X as a directed limit of Noetherian schemes.
- Descent of finite presentation and of closed immersions. Limits, Lemmas labelled limits-lemma-descend-finite-presentation and limits-lemma-descend-closed-immersion-finite-presentation, cited at proof lines 11098-11102. Descends Z -> X to an approximation.
- Base change stability of almost blowup squares. Same chapter, Lemma labelled lemma-base-change-almost-blow-up, cited at proof lines 11113-11115. Transports the Noetherian almost blowup square to X.

Source reference. Original equation 0EV6: sources/stacks-flat/pinned-original-source.tex, source lines 10957–10957. Same-chapter pointer into the author environment equation at source line 10957; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original proposition 01ZA: sources/stacks-chow/tags.txt, source lines 2585–2585. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 11097 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite located in the project tag list

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Source reference. Original lemma 01ZP: sources/stacks-chow/tags.txt, source lines 2602–2602. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 11101 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EV7: sources/stacks-flat/pinned-original-source.tex, source lines 10987–10987. Same-chapter pointer into the author environment lemma at source line 10987; invoked at proof line 11112 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let $Z \subset X$ be a closed subscheme cut out by a finite type quasi-coherent sheaf of ideals. Then there exists an almost blow up square as in (0EV6).*

Proof. We may write $X = \lim X_i$ as a directed limit of an inverse system of Noetherian schemes with affine transition morphisms, see Limits, Proposition 01ZA. We can find an index i and a closed immersion $Z_i \to X_i$ whose base change to X is the closed immersion $Z \to X$. See Limits, Lemmas 01ZM and 01ZP. Let $b_i : X_i' \to X_i$ be the blowing up with center Z_i. This produces a blow up square

$$\begin{array}{ccc} E_i & \longrightarrow & X_i' \\ \downarrow & & \downarrow{\scriptstyle b_i} \\ Z_i & \longrightarrow & X_i \end{array}$$

where all the morphisms are finite type morphisms of Noetherian schemes and hence of finite presentation. Thus this is an almost blow up square. By Lemma 0EV7 the base change of this diagram to X produces the desired almost blow up square. $\square$